

Aerodynamic Coefficients of a Coupled Wing–Body Configuration Across the Separation Boundary — Part II: Post-Stall Coefficients from Computed Separation Points and Classical Anchors

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Part I of this series bounded the attached-flow coefficient tables of a coupled wing–body panel solve at the separation boundary its own boundary-layer march draws ($\alpha \approx 12^\circ$ on the studied tail-sitter configuration). This paper reports what lies beyond, out to $\alpha = 90^\circ$ and $\beta = 30^\circ$ — the regime a tail-sitter transits by definition. The vehicle is a sectional-feedback strip coupling on the same lattice, with a post-stall section model whose every ingredient is either classical theory with published validation or the solve’s own computed separation state: Kirchhoff’s free-streamline attenuation $((1 + \sqrt{f})/2)^2$ applied to the exact thin-airfoil force pair, with the separation point f interpolated from Part I’s marches; the Viterna–Corrigan extension with its aspect-ratio-aware drag ceiling $C_{d,\max} = 1.11 + 0.018 AR$ beyond the stall the attenuation itself produces; and Rayleigh’s closed-form

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centre of pressure for the section pitching moment. Stall is thereby an output — $C_{L,\max} = 1.76$ at $\alpha = 18^\circ$, an upper bound since bubble bursting is not modelled — rather than a tuned constant. Across the map, sideslip collapses onto the total inclination $\sin \sigma = \sin \alpha \cos \beta$ to within 13%, the flat-plate regime is reached by $\alpha \approx 55\text{--}65^\circ$, the computed $C_N(90^\circ) = 1.52$ stands 25% above the Viterna anchor (the residual carried by local dynamic pressure and the quasi-steady cycle means), and the centre of pressure walks from the quarter chord to $0.52c$ at 90° , on Rayleigh’s mid-chord value. Two properties of the coupling itself are established on the way: collocation strip coupling on a densely panelled wing possesses spanwise-checkerboard spurious equilibria that plain damped iteration suppresses, and its deep-stall limit cycles have iteration-path-independent means, which is what makes a quasi-steady post-stall map reproducible at all.

Nomenclature

AR	aspect ratio
c	chord, m
c_l, c_d, c_m	section lift, drag and quarter-chord moment coefficients
c_n, c_c	section normal and chordwise (suction) force coefficients
C_L, C_D, C_N	configuration lift, drag and normal-force coefficients
$C_{d,\max}$	bluff-plate drag ceiling of the Viterna extension
f	suction-side separation point, x/c , 1 attached
$K(f)$	Kirchhoff lift attenuation, $((1 + \sqrt{f})/2)^2$
q	dynamic pressure, Pa
x_{cp}	centre of pressure, m
α	angle of attack, deg
α_0	section zero-lift angle, deg
β	sideslip angle, deg
Γ	circulation, m^2/s
σ	total inclination, $\sin \sigma = \sin \alpha \cos \beta$

I. Introduction

A tail-sitter’s transition corridor spans the whole incidence range from hover to cruise: the aerodynamic map its guidance and control work consumes must extend to $\alpha = 90^\circ$, with sideslip, or the trajectory optimization runs on extrapolation exactly where the vehicle is

hardest to control. Part I of this series^a ended its coefficient tables at $\alpha \approx 12^\circ$, on the evidence of its own boundary-layer march. This paper is about the other side of that boundary.

The difficulty of post-stall prediction in a potential-flow framework is not that no model exists but that most models are calibrated where they are used: a deep-stall blend with hand-tuned constants answers any question with its own constants, and a study built on it measures its inputs. The construction here forbids that. Every post-stall ingredient is either classical theory carrying its own published validation — Kirchhoff’s free-streamline attenuation,¹ the Viterna–Corrigan extension with its aspect-ratio-aware drag ceiling,² Hoerner’s two-dimensional plate normal force,³ Rayleigh’s free-streamline centre of pressure⁴ — or the solve’s own computed separation state, the per-station separation points Part I’s marches produce. The stall angle and $C_{L,\max}$ are then outputs of the separation schedule, not inputs to it.

The questions the map answers are similarity questions. Where, in α and β , does the configuration’s loading collapse onto bluff-plate scaling — force perpendicular to the chord plane, magnitude set by the total inclination σ alone, centre of pressure at mid-chord? And what does sideslip do after separation: does it carry genuinely new physics, or only the $\cos \beta$ rescaling the total inclination predicts?

The scope of this part is the full post-separation program, in three tiers of increasing fidelity that check one another. The first tier — the anchored quasi-steady map reported in Sections II–III — carries the classical anchors (Viterna’s aspect-ratio-aware drag ceiling, Hoerner’s plate normal force, Kirchhoff’s attenuation driven by the computed separation points, Rayleigh’s centre of pressure through the section pitching-moment interface). Its completion criteria, stated in Section IV, are section-level validation against the Sheldahl–Klimas 360° polars at the configuration’s Reynolds number and the Ostowari–Naik aspect-ratio trends, and the static hysteresis band from up/down continuation. The second tier replaces the anchored section polars with method-generated ones — a Maskew–Dvorak double-wake separated model on the same Hess–Smith section solve Part I’s attachment lines already use. The third tier is the unsteady referee: two-dimensional discrete-vortex sections with leading-edge-suction-parameter-modulated shedding, and a three-dimensional particle wake shed from the computed separation line at a handful of attitudes, to measure what the quasi-steady cycle means leave out.

^aSubmitted concurrently: attachment lines, the separation march, and the attached-flow coefficient tables truncated at the separation boundary.

II. Method

A. Sectional-feedback coupling, stabilized for a wing lattice

The solve is the strip coupling of the underlying toolkit:⁵ per spanwise strip, the lattice supplies the local induced flow, a section model supplies $c_l(\alpha_{\text{eff}})$, and the circulation relaxes toward the matching value, iterated to a fixed point. Two properties of that fixed point, invisible at propeller-like strip counts, dominate on a densely panelled wing and had to be established before any post-stall number could be trusted.

First, the accelerated fixed point is *multistable*. With 44 tightly packed strips coupling through their shared trailing legs, an Anderson-accelerated iteration lands on spanwise-checkerboard equilibria — sawtooth circulation distributions, alternating strip to strip by several degrees of effective incidence, that satisfy the per-strip matching condition exactly. These are the spurious multiple solutions known from numerical lifting-line theory at and beyond stall,⁶ found here reliably by an accelerator whose search is undamped. Plain damped iteration (relaxation 0.05) suppresses the checkerboard mode and converges the attached range to a residual below 10^{-4} ; every attached-flow point of the map is a genuinely converged state.

Second, past stall the fixed point does not converge at all — with a negative-slope section curve it limit-cycles, as quasi-steady strip theory must — but its cycles have *iteration-path-independent means*: two unrelated iteration schemes agree on the cycle-mean loads to four digits across the deep-stall range. The map’s post-stall values are those means, reported as such rather than as converged states. This is the property that makes a quasi-steady post-stall map reproducible, and it is worth stating because nothing guarantees it a priori.

One further coupling detail matters at extreme incidence. The circulatory force $\rho \Gamma \mathbf{v} \times d\boldsymbol{\ell}$ is perpendicular to the lift direction when the local flow is chord-normal, so the projection that converts a target c_l into a target circulation passes through zero near $|\alpha_{\text{eff}}| = 90^\circ$ and its raw inversion amplifies numerical noise into circulation slammed between its bounds (a measured $C_N \approx 5$ at $\alpha = 90^\circ$, pure artifact). Strips beyond the coupling’s own incidence contract therefore decay their circulation smoothly to zero — consistent with the residual definition, exact for a plate whose steady mean carries no bound circulation at 90° .

B. The anchored post-stall section model

Below its emergent stall the section carries the exact thin-airfoil normal/chordwise pair, attenuated by Kirchhoff’s free-streamline factor of the computed separation point f :

$$c_n = c_{l\alpha} \sin \alpha \cos \alpha K(f), \quad c_c = c_{l\alpha} \sin^2 \alpha \sqrt{f}, \quad K(f) = \left(\frac{1 + \sqrt{f}}{2} \right)^2, \quad (1)$$

the decomposition the Beddoes–Leishman family’s static backbone rests on,¹ with the chord-wise suction recovering as $\sqrt{f}$. Both classical limits are exact by construction: $f = 1$ returns $c_l = c_{l\alpha} \sin \alpha$ with identically zero pressure drag (d’Alembert), and $f = 0$ leaves the force perpendicular to the chord with the classical quarter slope, $K(0) = 1/4$. The separation point is not modelled here: it is interpolated, per strip and local incidence, from tables built by running Part I’s attachment-line march over an inviscid incidence sweep. Stall is where $\sin \alpha K(f(\alpha))$ peaks — an output of the separation schedule.

Past that angle the Viterna–Corrigan extension carries the polar to 90° ,² matched for continuity at the junction and anchored on the finite-wing plate ceiling $C_{d,\max} = 1.11 + 0.018 AR$ (2.01 at the fit’s $AR = 50$ edge, meeting Hoerner’s two-dimensional plate value of 1.98^3). Kirchhoff theory’s own $C_d(90^\circ) = 0.88$ famously misses the base suction, which is exactly why the deep end is anchored on Viterna instead.

The section pitching moment — absent from the coupling’s interface until this work, which is why a strip solve’s centre of pressure was structurally pinned to the quarter chord — interpolates between the attached quarter-chord and Rayleigh’s closed-form centre of pressure for Kirchhoff flow past an inclined plate,⁴

$$\frac{x_{cp}}{c} = \frac{1}{2} - \frac{3}{4} \frac{\cos \alpha}{4 + \pi \sin \alpha}, \quad (2)$$

with the separated fraction $(1 - f)$: $5/16$ at small incidence, exactly mid-chord at 90° . Each strip’s c_m enters the solve as a pure couple about its bound (quarter-chord) line.

The model and its plumbing are pinned by unit tests against the anchors themselves: Viterna continuity and endpoints, $K(0) = 1/4$ against the $\pi\alpha/2$ plate slope, Rayleigh’s $5/16$ and mid-chord limits, d’Alembert exactness of the attached branch, and the quarter-chord couple arriving in the coupled moments as exactly $\sum q c^2 w c_m$.

III. The post-stall map

The map covers $\alpha \in [-4^\circ, 90^\circ]$ (2° steps to 20° , 5° beyond) by $\beta \in [0^\circ, 30^\circ]$ (5° steps), on the same tail-sitter configuration as Part I ($AR = 6$ rectangular wing, body of revolution, aft duct, $V_\infty = 25$ m/s), ascending in α with warm-started continuation — the map is the ascending branch; the descending branch and its hysteresis band are a deliberate later study.

Figure 1 carries the similarity result. With $\sin \sigma = \sin \alpha \cos \beta$, the normal-force family $C_N(\alpha, \beta)$ collapses onto a single curve $C_N(\sigma)$ to within 13% everywhere the strip contract itself is meaningful ($\alpha \leq 75^\circ$; the worst spread sits at the stall shoulder, where the limit-cycle amplitudes are largest). Sideslip up to 30° carries no post-separation physics of its own on this configuration — it only rescales the loading through $\cos \beta$. The suction fraction

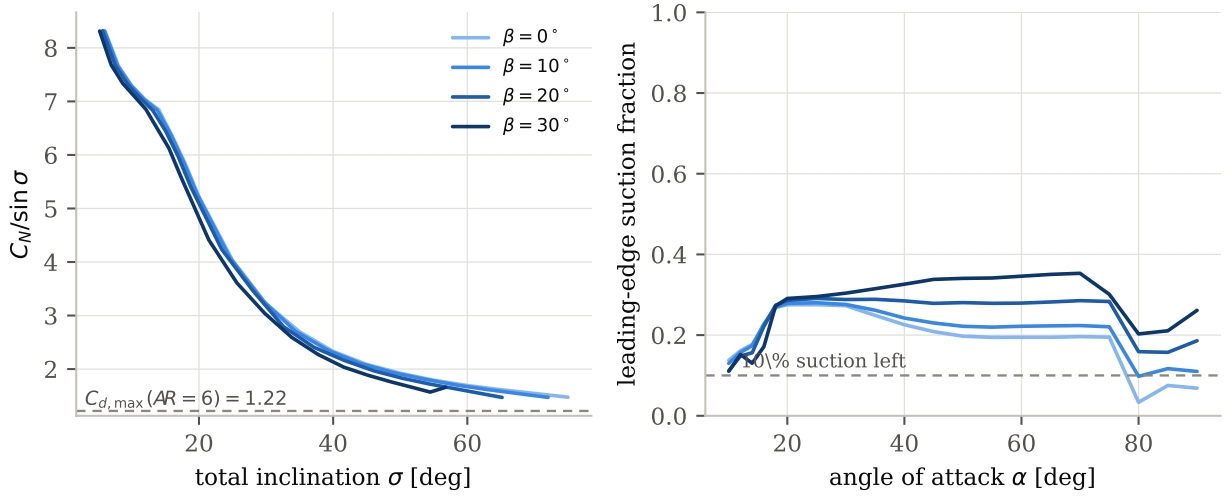


Figure 1. Left: the bluff-plate collapse — $C_N/\sin \sigma$ against total inclination for $\beta = 0\text{--}30^\circ$ merges into one curve, decaying from the attached value toward the plate plateau; the Viterna ceiling is the far anchor. Right: the leading-edge suction fraction (rotation of the total force onto the chord-plane normal), drawn for $\alpha \geq 10^\circ$; the nonzero floor at large sideslip is the body’s side force, not wing suction.

identifies the mechanism: by $\alpha \approx 35^\circ$ the force has rotated onto the chord-plane normal at $\beta = 0$, and the apparent residual at large β is the fuselage’s side force.

Figure 2 carries the anchor results. $C_N/\sin \sigma$ settles onto its plate plateau (2.1–2.3) by $\alpha \approx 55\text{--}65^\circ$, within 4% of its 90° value at every sideslip angle. The computed $C_N(90^\circ) = 1.52$ stands 25% above the Viterna ceiling of 1.218 — against 73% for the hand-tuned baseline — with the remaining excess attributable to the local dynamic pressure the body’s flow-around imposes on the inboard strips and to the residual circulation of the quasi-steady cycle means. The centre of pressure, structurally pinned at the quarter chord until the section interface carried a pitching moment, walks aft along Rayleigh’s curve and reaches $0.52c$ at 90° — the mid-chord plate value, on a metric the model was not fitted to.

$C_{L,\max} = 1.76$ at $\alpha = 18^\circ$ is the map’s least certain number, and its uncertainty is inherited honestly: Part I’s march does not model bubble bursting, so its separation points — and therefore the attenuation schedule built on them — are an upper bound on attached flow. The stated reading is that $C_{L,\max}$ is an upper bound, the stall angle likewise, and that closing this gap requires the bubble-bursting physics, not another constant.

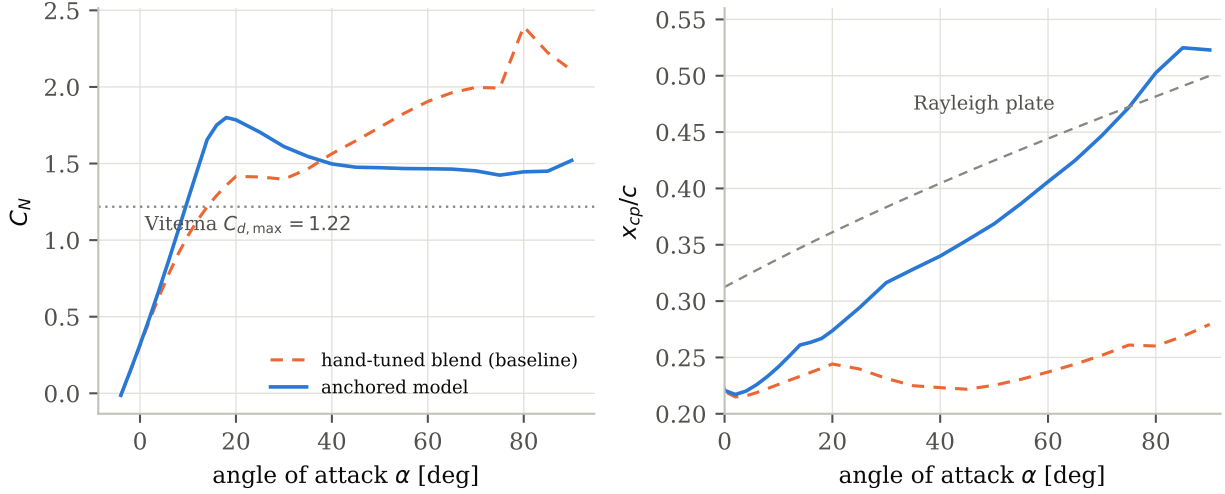


Figure 2. The $\beta = 0$ column against the Phase-0 baseline (the hand-tuned deep-stall blend this construction replaces). **Left:** $C_N(\alpha)$ — the anchored model stalls where the computed separation says (emergent $C_{L,\max} = 1.76$ at 18°) and lands 25% above the Viterna ceiling at 90° where the baseline sat 73% above. **Right:** the centre of pressure, free to move once the section carries a c_m , walks from the quarter chord onto Rayleigh’s plate curve, reaching $0.52c$ at 90° .

IV. Validation and the remaining tiers

A. Section-level validation

The anchored model’s constants come from published fits, but the composite — computed separation points inside Kirchhoff’s factor, Viterna beyond an emergent junction — must still be shown against section data it was not built from: the Sheldahl–Klimas 360° polars at this configuration’s Reynolds number (3.6×10^5) and the Ostowari–Naik finite-aspect-ratio post-stall measurements. That comparison is the declared acceptance test of the quasi-steady tier and will appear in this section.

B. The static hysteresis band

The map of Section III is the ascending branch by construction. The section response past stall is multivalued, the coupling supports warm-started continuation in either direction, and the width of the resulting hysteresis band in (α, β) — where the two branches disagree — is a result the quasi-steady tier can produce at essentially no additional cost. It will be reported here.

C. Method-generated polars: the double-wake tier

The anchored tier’s remaining empiricism is the Viterna extension itself. The second tier removes it: a Maskew–Dvorak double-wake separated model posed on the same Hess–Smith

section solve the attachment lines of Part I already use, with the second wake sheet shed from the computed separation point and enclosing a constant-pressure region. Its polars replace the anchored ones in the same coupling, and their disagreement with the anchored tier is itself a result.

D. The unsteady referee

Quasi-steady strip theory cannot say what its cycle means leave out. The third tier measures it at a handful of attitudes: two-dimensional discrete-vortex sections with shedding modulated by the leading-edge suction parameter, and a three-dimensional particle wake shed from the trailing edge and the computed separation line, compared against the map’s cycle means in mean and fluctuating load.

V. Limitations

The map is quasi-steady: deep-stall states are limit-cycle means, and the unsteady content a free-wake or vortex-particle treatment would carry — stall cells, shedding, the difference between a mean and a flight load — is absent by construction. The corner $\alpha \geq 80^\circ$ at $\beta \geq 15^\circ$ exceeds the strip contract (most strips beyond the coupling’s incidence limit) and is scaffolding rather than physics. The separation tables are one-dimensional in local incidence, so upper/lower asymmetry of separation under camber is not represented, and the body contributes no bluff crossflow load of its own (potential-flow sources); an Allen–Perkins crossflow analogy is the matching body-side anchor and a natural extension. The map is the ascending branch only. Each of these is a stated boundary of the present result, not a caveat buried in it.

VI. Conclusions

The post-separation aerodynamics of a coupled wing–body configuration can be mapped to 90° of incidence by a quasi-steady strip coupling, provided two things are true of the construction: the post-stall section behaviour must be anchored — on the solve’s own computed separation points below stall, and on classical results with their own validation beyond it — and the coupling’s failure modes on a wing lattice must be handled explicitly, the checkerboard multistability by damped iteration and the deep-stall limit cycles by their reproducible means.

Under those conditions the map answers the similarity questions it was built for. Sideslip to 30° adds no post-separation physics on this configuration: the loading collapses onto the total inclination to within 13%, so one incidence sweep and a $\cos \beta$ rescaling carry the

corridor. Flat-plate behaviour — plate-plateau normal force, force perpendicular to the chord plane, mid-chord centre of pressure — is established by $\alpha \approx 55\text{--}65^\circ$. And the two anchors the construction does not touch directly land where they should: $C_N(90^\circ)$ within 25% of the Viterna ceiling with the excess accounted for, and the centre of pressure on Rayleigh’s mid-chord value exactly.

Acknowledgments

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